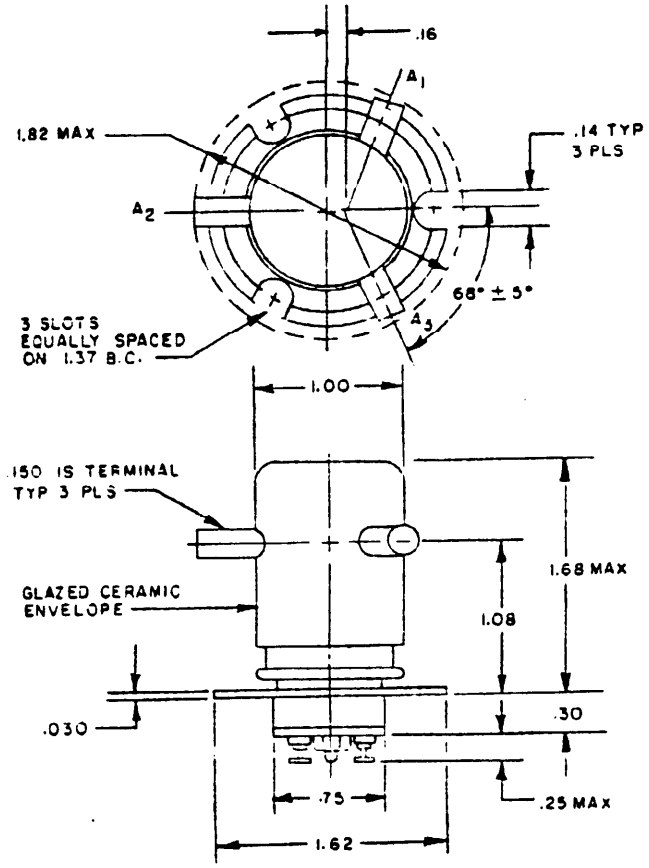
MILITARY SPECIFICATION SHEET

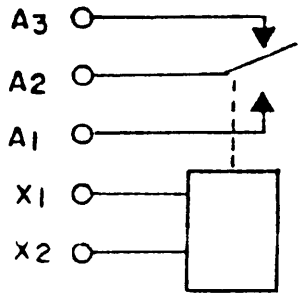
RELAYS, VACUUM, SPDT, 30 AMPERES RMS, 15 KILOVOLTS (PEAK)

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relays described herein shall consist of this specification and the latest issue of MIL-R-83725.



Inches	mm
.030	0.76
.14	3.6
.150	3.81
.16	4.1
.25	6.4
.30	7.6
.75	19.1
1.00	25.4
1.08	27.4
1.37	34.8
1.62	41.1
1.68	42.7
1.82	46.2



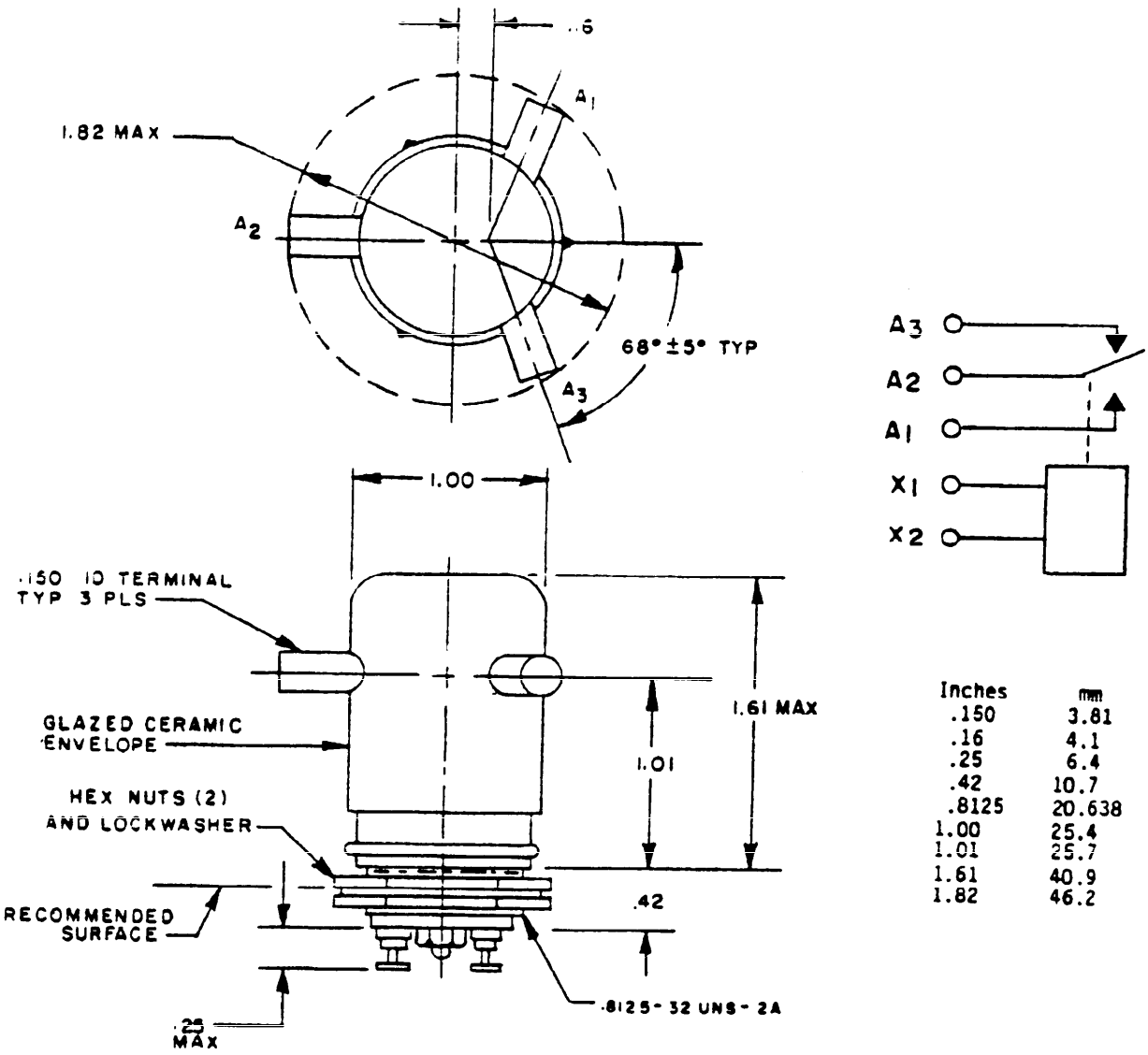
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.8 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
4. Terminal identification and locations designated by diagram on relay case.
(A) 5. Marking shall be in accordance with MIL-STD-1285.

(A) FIGURE 1. Relay outline drawing (flange mount).

(A) denotes changes.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Inches	mm
.150	3.81
.16	4.1
.25	6.4
.42	10.7
.8125	20.638
1.00	25.4
1.01	25.7
1.61	40.9
1.82	46.2

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.03 (0.8 mm) for two place decimals and ±.010 (0.25 mm) for three place decimals.
4. Terminal identification and locations designated by diagram on relay case.
Ⓐ 5. Marking shall be in accordance with MIL-STD-1285.

Ⓐ FIGURE 2. Relay outline drawing (thread mount).

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REQUIREMENTS:

Duty rating: Continuous.

Contact data:

Configuration: SPDT.

Arrangement: 1 form C.

Load ratings: (carry).

Resistive: 30 amperes rms, dc or 60 hertz.

Voltage ratings: 1 kilovolt peak, dc or 60 hertz.

Contact resistance, rated life:

Before: .025 ohm maximum.

During: .200 ohm maximum.

After: .200 ohm maximum.

Contact bounce: Not applicable.

Capacitance:

1.5 picofarads maximum across open contacts.

2.0 picofarads maximum between open contacts and ground.

Coil data: (See table I).

Operate time: 18 milliseconds maximum, including bounce.

Release time: 9 milliseconds maximum, including bounce.

ELECTRICAL DATA:

Insulation resistance: 1,000 megohms minimum, except the resistance between coil and housing at high temperature shall be 500 megohms or greater.

Dielectric withstanding voltage:

At atmospheric pressure:

Between all mated contacts in open position: 17 kilovolts peak, 60 hertz.

Between high voltage terminals and housing: 17 kilovolts peak, 60 hertz.

Between coil and housing: 500 volts rms, 60 hertz.

At reduced barometric pressure: In accordance with MIL-STD-202, method 105, test condition C. Relays shall be mounted by normal means. Points of application - Same as "at atmospheric pressure" except test voltage applied shall be 350 volts rms with all terminals other than test terminal grounded. Following the test, relays shall be examined for evidence of arcing, flashover, insulation breakdown, and damage.

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ENVIRONMENTAL DATA:

Temperature range: -55°C to +125°C.

Vibration: MIL-STD-202, method 204, test condition A, 10 G, 10-500 Hz.

Shock: MIL-STD-202, method 213, test condition J.

PHYSICAL:

Terminal strength: 5 pounds pull.

Dimensions and configurations: See figures 1 and 2.

Weight: 4 ounces maximum.

LIFE TEST REQUIREMENTS:

Mechanical cycling: 100,000 operations. Two sample units (maximum cycling rate, 5 hertz; contact current shall not exceed 10 milliamperes).

QUALITY ASSURANCE:

Dielectric withstanding voltage: (Tests to be conducted at atmospheric pressure rating only).

Duration of application: 5 to 10 seconds at a 10-percent increase in the dielectric withstanding voltage.

PART NUMBER: M83725/20- (dash number from table I).

TABLE I. Coil data and dash numbers.

Dash number M83725/20-	Figure	At +20°C						Over temperature range		
		Coil voltage nominal (V dc) 1/	Coil resistance (Ohms ±10%)	Coil voltage maximum (V dc)	Pickup voltage maximum (V dc)	Dropout voltage (V dc)		Pickup voltage over temperature range maximum (V dc)	Dropout voltage over temperature range (V dc)	
						Min	Max		Min	Max
001	1	12	48	14	8	0.5	5	11	0.3	7
002	1	26.5	180	32	16	1	10	23	0.7	14
003	1	115	2900	127	80	5	50	110	3.5	70
004	2	12	48	14	8	0.5	5	11	0.3	7
005	2	26.5	180	32	16	1	10	23	0.7	14
006	2	115	2900	127	80	5	50	110	3.5	70

1/ CAUTION: Use of any coil voltage less than the nominal voltage will compromise the operation of the relay.

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CONCLUDING MATERIAL

Custodian:
Air Force - 85

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-F686)

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